Week 9

Class schedule for the rest of the semester

	Lecture	**What is due on that day?**
November 10th	SVM	
November 17th	Review, GenAI – images	
November 24th	GenAI – text	HW 4 due
December 1st	Exam 2	All of the material past Exam 1 – 1 double sided cheat sheet
December 8th	Final Project Presentation	Final Project Due

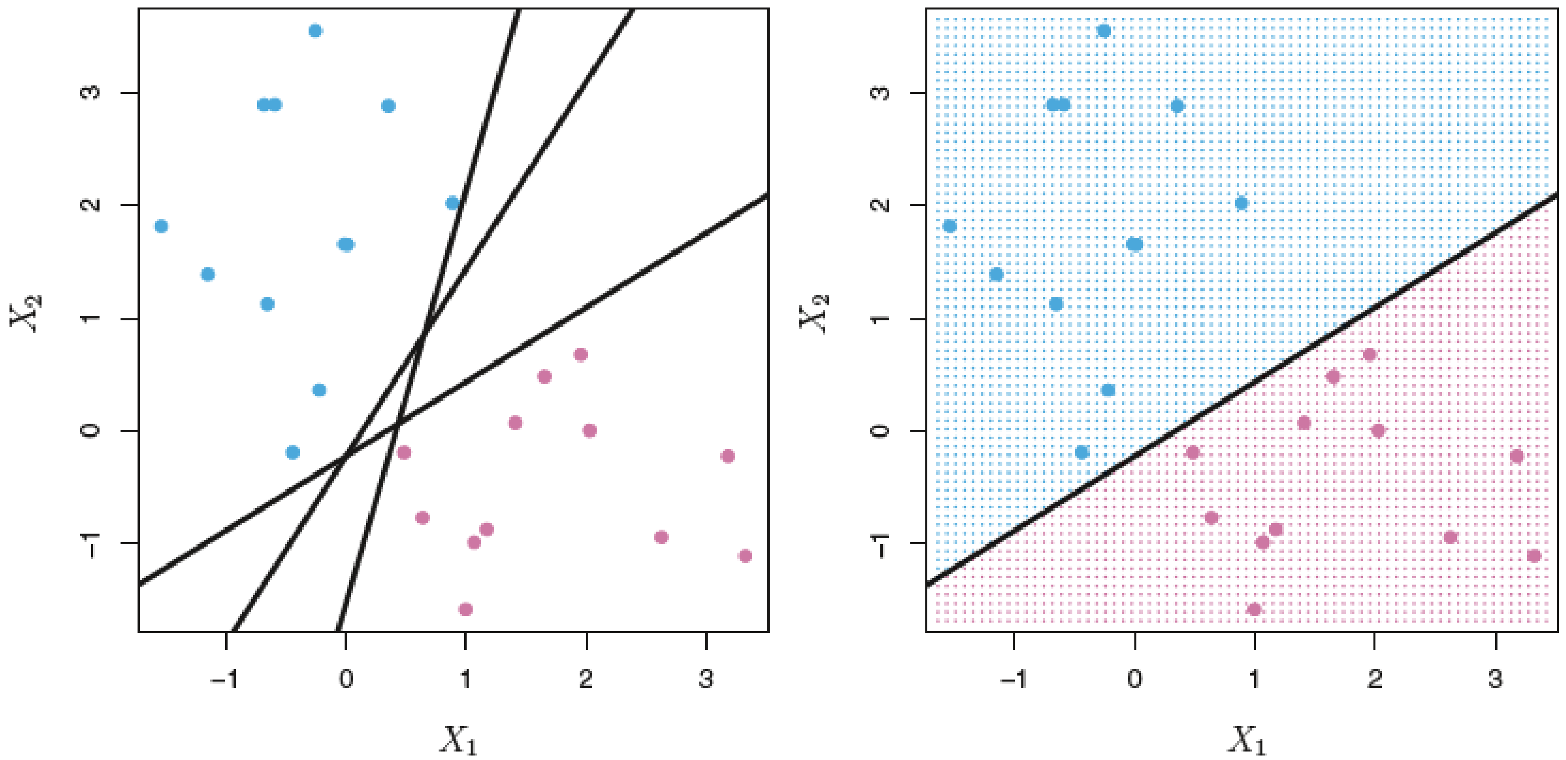
Agenda

- SVN
- Pushing stuff to GitHub

Support Vector Machines for Classification Problems

Support Vector Machines (SVM) finds a decision boundary between two classes just like logistic regression, but it selects the boundary with the widest possible distance between two classes.

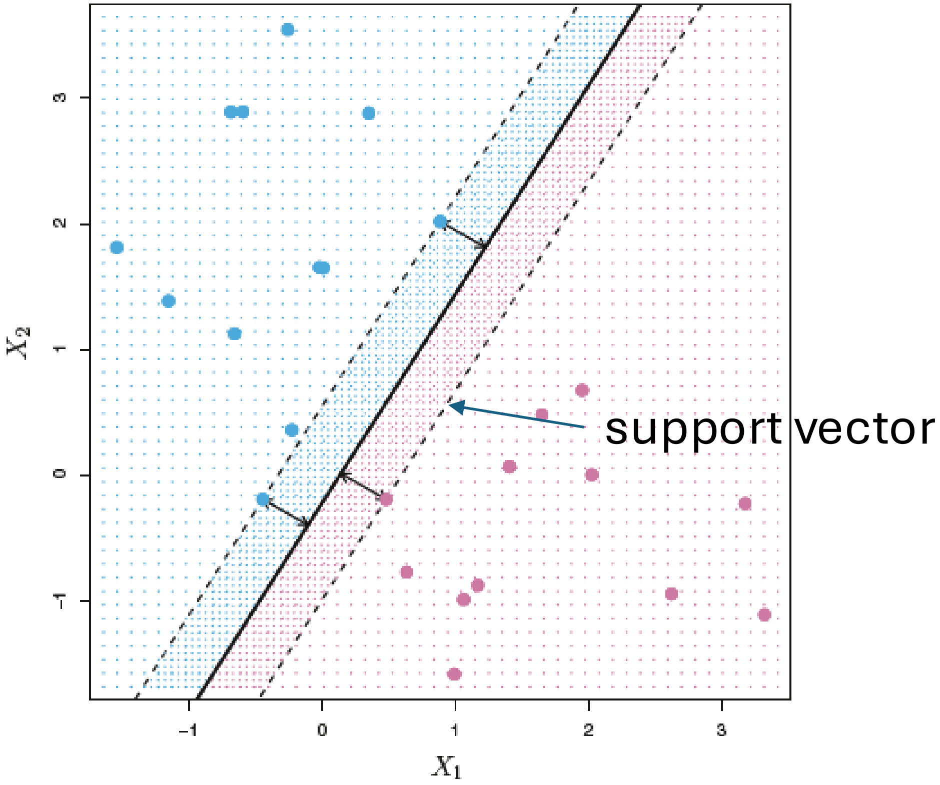
Decision Boundaries



- Notice we can have many decision boundaries in a space
- We want the decision boundary that will best separate our outcome into 2 categories

Decision Boundary for SVM

Among all separating boundaries, find the one that makes the biggest gap or margin between the two classes.



The decision boundary is found using Euclidean distance.

The space between the boundary and the points closet to it is called the ***margin***.

Find the boundary that separates the classes with the widest margin. The points that lie on the margin are the ***support vectors***.

Pro Tip: Feature Scale

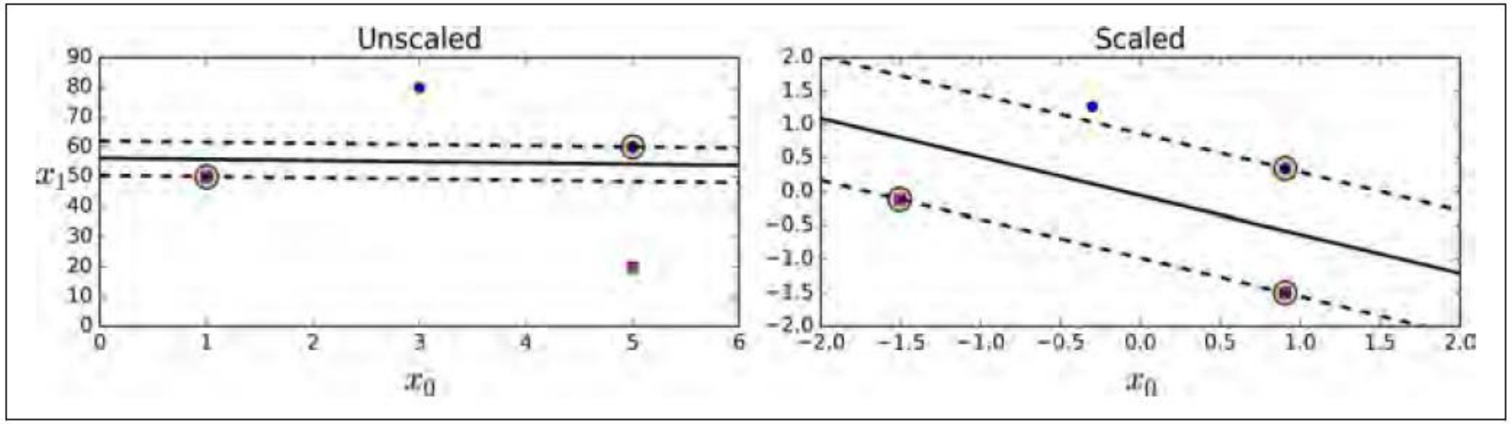
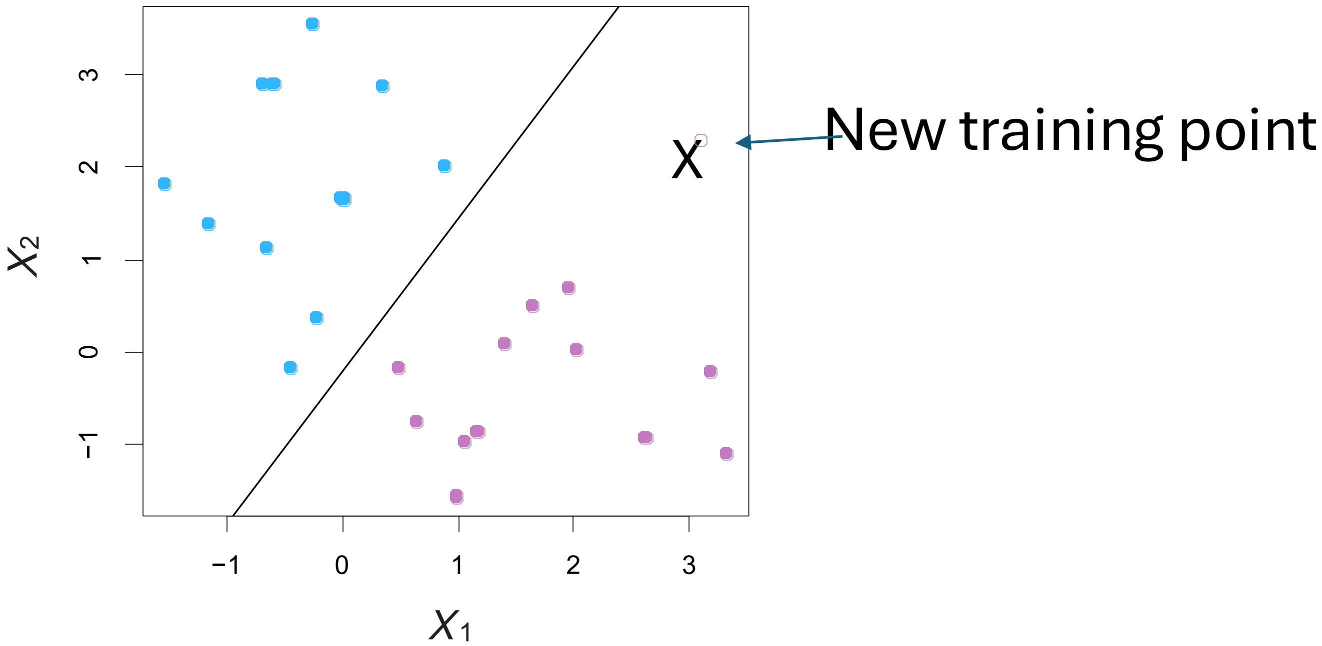


Figure 5-2. Sensitivity to feature scales

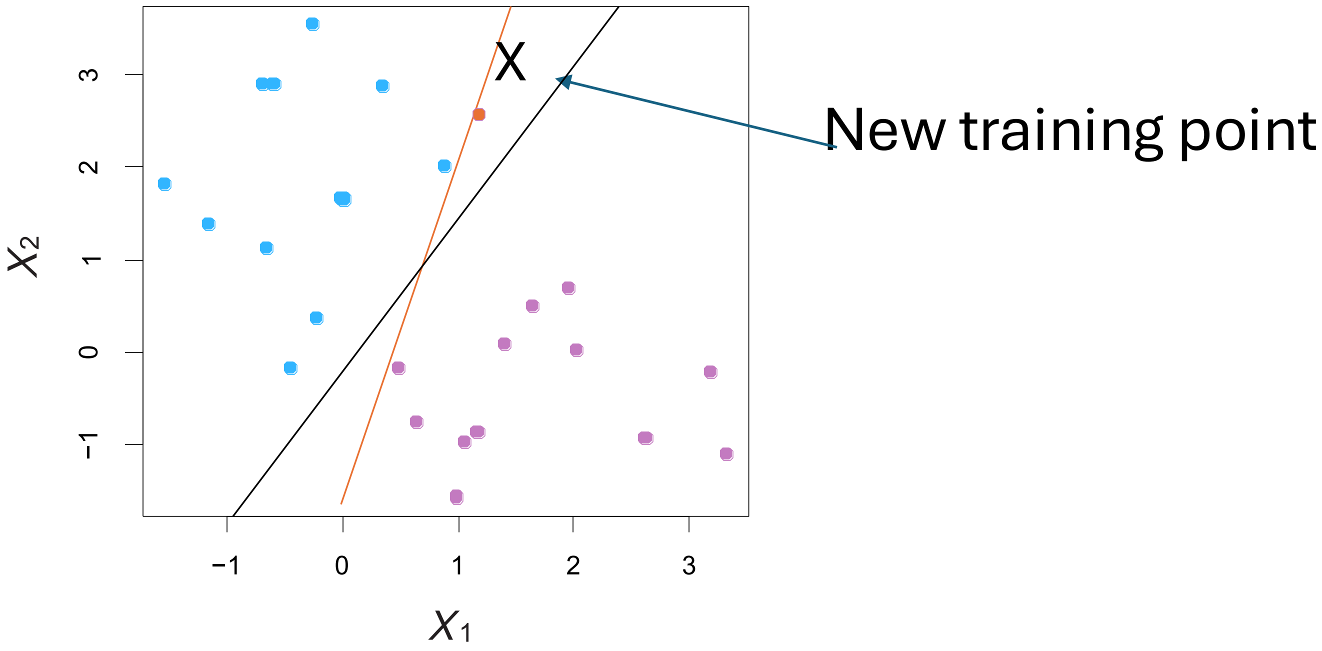
It is best to normalize features when using SVMs. You will get a wider decision boundary.

Outliers



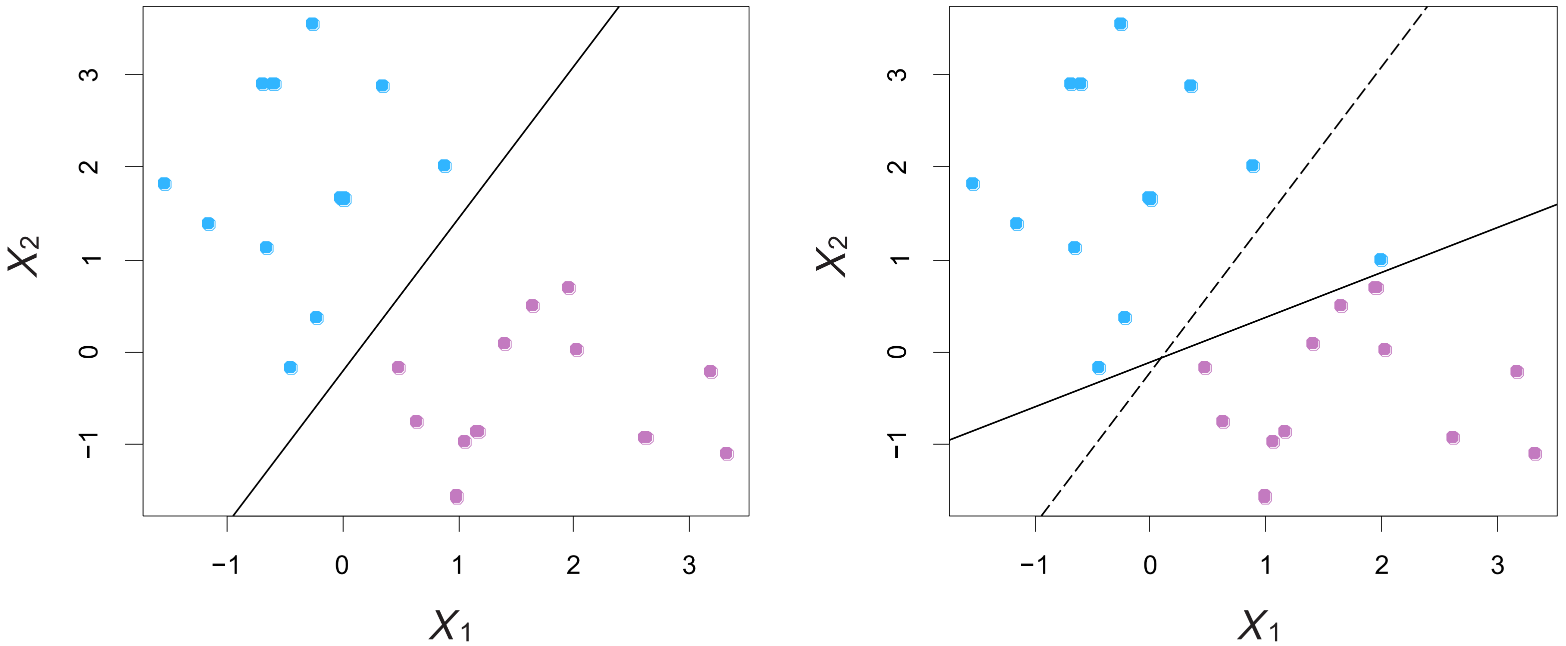
Once a hyperplane is found, adding training data that would affect other machine learning models won't affect the SVM, as long as the new training point is on the correct side of the decision boundary.

Outliers



But a new point that lies on the other side of the hyperplane does affect it. We have to find a new hyperplane.

Noisy Data



Sometimes the data are separable, but noisy. This can lead to a poor solution.

The *support vector classifier* maximizes a *soft* margin.

Hard vs Soft Classification

Hard margin classification means that all instances of the same class must be on one side of the decision boundary.

Issues:

1. Only works if data is linearly separable
2. It is sensitive to outliers

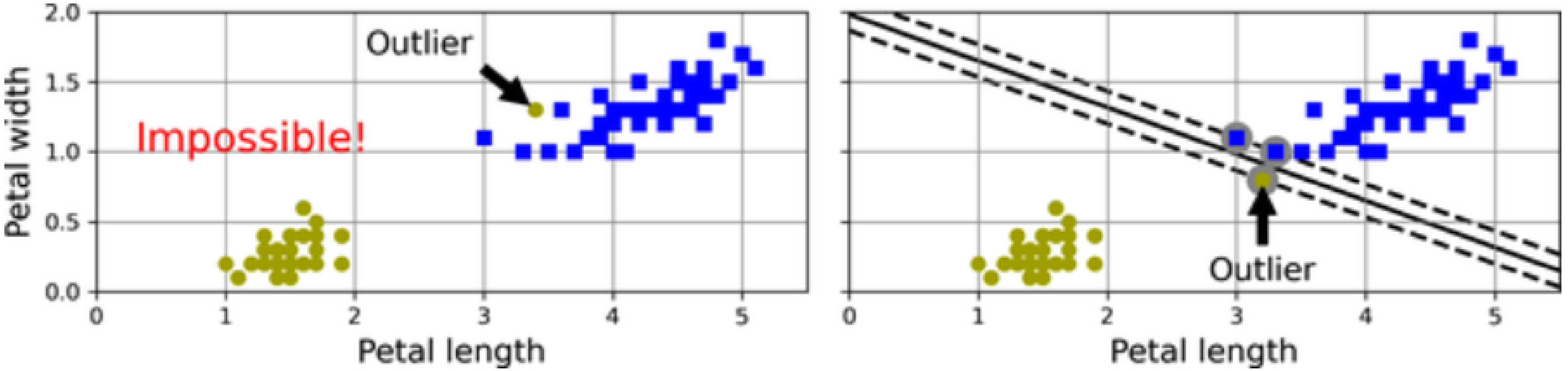


Figure 5-3. Hard margin sensitivity to outliers

Soft Margin

The soft margin allows us to specify a parameter c where c is the number of instances allowed within the margin or on the wrong side of the decision boundary.

The key is to finding the optimal value for c.

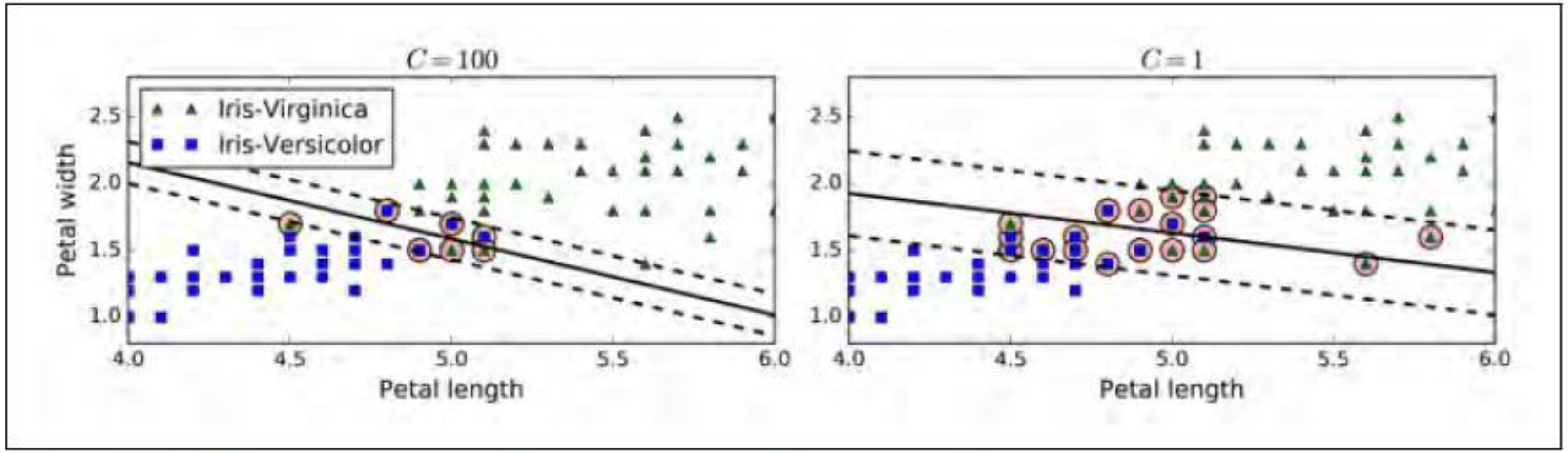


Figure 5-4. Fewer margin violations versus large margin

Hyperparameter

C is referred to as a ***hyperparameter***, *a parameter whose value is set to control the learning process. It is set before the learning takes place and can be altered to fine tune the learning model*.

A hyperparameter is not found mathematically by an algorithm.

Hyperparameters are flexible and the model benefits by varying the hyperparameters values.

Hyperparameter Examples

Examples of hyperparameters:

The number of epochs/iterations and the learning rate used in gradient descent.

The max depth and minimum samples leaf of a decision trees.

The number of bags in a bag classifier.

The number of features used in best feature selection in a random forest.

Choosing the best hyperparameters for a model makes a huge difference in the accuracy of a model.

What sklearn class can we use to help with this? *GridSearchCV*

Linear boundary can fail

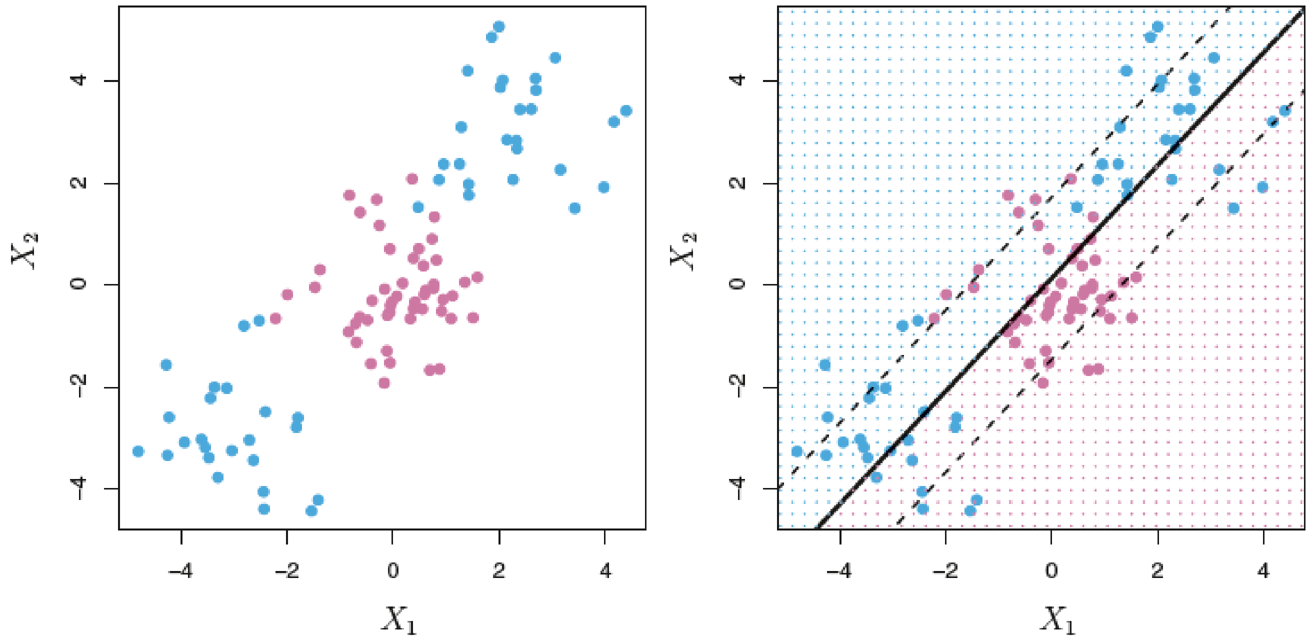


FIGURE 9.8. Left: *The observations fall into two classes, with a non-linear boundary between them.* Right: *The support vector classifier seeks a linear boundary, and consequently performs very poorly.*

Sometimes a linear boundary simply won't work, no matter what value of c is chosen.

What to do? – use feature expansion

Feature Expansion

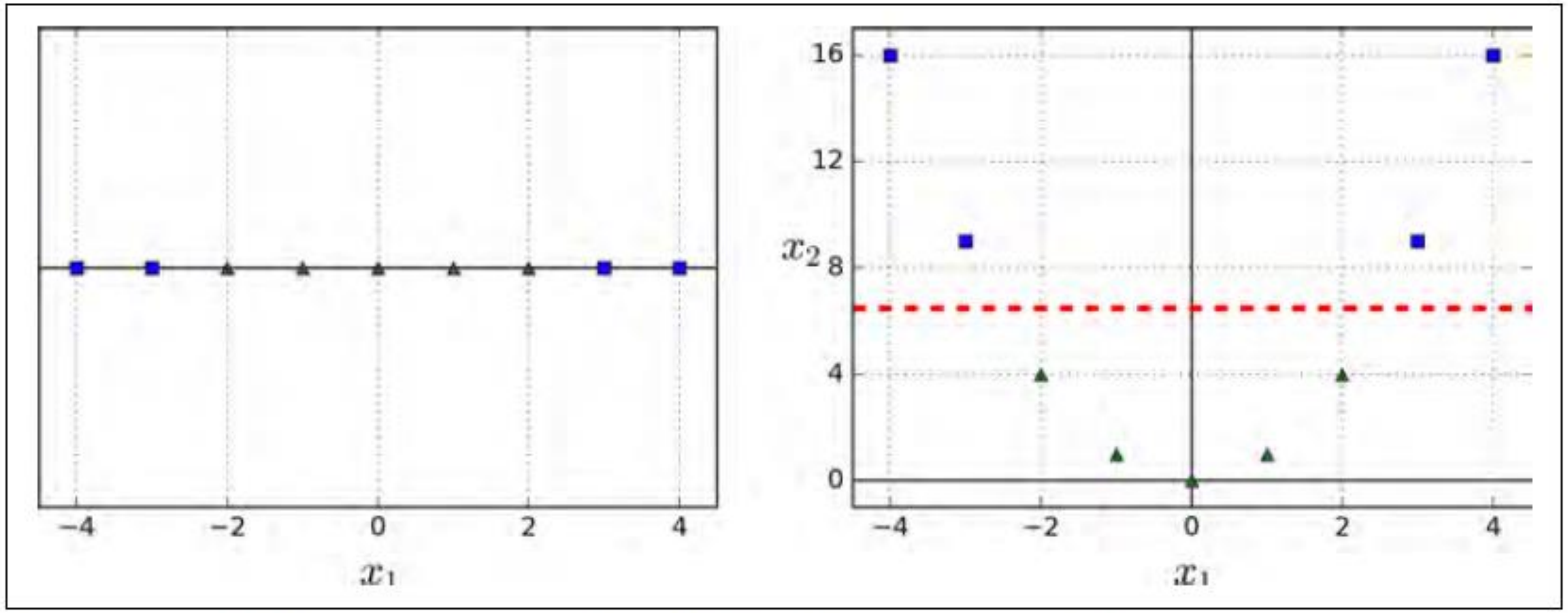


Figure 5-5. Adding features to make a dataset linearly separable

The graph on the left shows the decision boundary for a data set with one feature, X_1. It is not linearly separable.

If we add a polynomial feature X_2 that is X_1^2, our results are now linearly separable.

Feature Expansion

- Enlarge the space of features by adding polynomial features.
- Fit a support-vector classifier in the enlarged space.
- This results in non-linear decision boundaries in the original space.
- Note: the decision boundary is linear in the M dimensional space but non-linear in the original space.

Github

Pull = download latest changes from remote → your local machine
Push = upload your committed changes → remote GitHub repo
•Always **pull before you start working**

```
git clone <repo-url>          # download repo first time
git pull                      # update your local branch
git add .                     # stage changes
git commit -m "message"       # save locally
git push                      # upload to GitHub
```

Github

```
git branch feature/my-branch      # create a branch
git checkout feature/my-branch    # switch
git push -u origin feature/my-branch
```

Why Branches?

- Isolate work without touching main/dev
- Safe experimentation
- Parallel development

Common Workflow

- **Main** → production-ready
- **Dev** → integration/testing
- **Feature branches** → short-lived (feature/fix-logging)

Github

What GitHub Actions Do?

- Run automated workflows on:
  - push
  - pull request
  - schedule
- Ensure **tests pass, code is clean, build succeeds** before merge
- YALM files

```yaml
name: CI

on: [push, pull_request]

jobs:
  build:
    runs-on: ubuntu-latest
    steps:
    - uses: actions/checkout@v3

    - name: Set up Python
      uses: actions/setup-python@v4
      with:
        python-version: "3.10"

    - name: Install deps
      run: pip install -r requirements.txt

    - name: Run tests
      run: pytest --maxfail=1 --disable-warnings

    - name: Lint
      run: ruff check .
```